WINC1500 Software

Release Notes

VERSION : **19.7.11**

DATE : **FEB, 2026**

Abstract

This document presents an overview of the WINC15x0 firmware release version 19.7.11, and corresponding driver.

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1 Introduction

This document describes the WINC15x0 version 19.7.11 release package. This is a release built on top of 19.7.10 with critical bug fixes and enhancements.

The release package contains all the necessary components (binaries and tools) required to make use of the latest features including tools, driver code and firmware binaries.

1.1 Highlights of the release

- Critical security fix:
    - TLS root certificate validation bypass vulnerability introduced in 19.7.10, where self-signed certificates were accepted without verification (W1500-984)
- Fixes for a number of important issues:
    - An OTA issue that could result in the WINC1500 erroneously switching to an invalid image (W1500-974)
    - The WINC1500 DHCP client sending an empty host name (option 12), which could prevent IP address assignment from strict DHCP servers (W1500-977)
    - Multiple OTA robustness improvements including better error reporting and socket conflict prevention (W1500-979, W1500-980)
    - MAC address handling now uses a MAC address in efuse if it looks valid even if the "MAC Address Used" bit is not set (W1500-978)
    - WPA2 Enterprise connection attempts could sometimes take up to 30 seconds due to dropped EAP Identity Request frames (W1500-327)
    - TLS Server mode with ECDHE-RSA ciphersuites no longer requires an ECDSA certificate chain (W1500-852)
    - All mandatory MISRA C coding standard warnings resolved in ASF driver code (W1500-990)

1.2 Firmware readiness

Microchip Technology Inc. considers version 19.7.11 firmware to be suitable for production release.

2 Release summary

2.1 Auditing information

Master Development Ticket	:	https://jira.microchip.com/projects/W1500/versions/88786
Release Repository	:	wsg_svn_wifi_m2m
Source Branch	:	release/rel_1500_19.7.11
Git Commit	:	f131715f

2.2 Version information

WINC Firmware version	:	19.7.11
Host Driver version	:	19.7.11
Minimum driver version	:	19.3.0

2.3 Released components

The release contains documentation, sources and binaries.

2.3.1 Documentation overview

The Application manuals, Release notes and Software API guides can be found in the `doc/` folder of the release package.

Release Notes:

This document

Software APIs:

WINC1500_IoT_SW_APIs.chm

2.3.2 Binaries and programming scripts

The main WINC15x0 firmware binary is located in the `firmware` directory and named `m2m_aio_3a0.bin`. This can be flashed to a WINC device using, for example, a serial bridge application available from ASF.

An OTA image is provided in the `ota_firmware` directory named `m2m_ota_3a0.bin`.

2.3.3 Sources

Source code for the host driver can be found under the `src/host_drv` directory.

Source code for the tools, including crypto_lib, can be found under the `src/Tools` directory.

2.4 Release Comparison

Features in 19.7.10	Changes in 19.7.11
Wi-Fi STA	
• IEEE 802.11 b/g/n. • OPEN security. • WPA Personal Security (WPA1/WPA2). • WPA Enterprise Security (WPA1/WPA2) supporting: EAP-TTLSv0/MS-Chapv2.0 EAP-PEAPv0/MS-Chapv2.0 EAP-PEAPv1/MS-Chapv2.0 EAP-TLS EAP-PEAPv0/TLS EAP-PEAPv1/TLS • 'Fragattack' countermeasures	No change
Wi-Fi Hotspot	
• Only ONE associated station is supported. After a connection is established with a station, further connections are rejected. • OPEN and WPA/WPA2 security modes. • The device cannot work as a station in this mode (STA/AP Concurrency is not supported). • 'Fragattack' countermeasures.	No change
WPS	
The WINC1500 supports the WPS protocol v2.0 for PBC (Push button configuration) and PIN methods.	No change
TCP/IP Stack	
The WINC1500 has a TCP/IP Stack running in firmware side. It supports TCP and UDP full socket operations (client/server). The maximum number of supported sockets is currently configured to 11 divided as: • 7 TCP sockets (client or server). • 4 UDP sockets (client or server).	No change
TLS	
• Support TLS v1.2. • Client and server modes. • Mutual authentication in client mode.	No change

Features in 19.7.10	Changes in 19.7.11
• Subject Alternate Name support in client mode • X509 certificate revocation scheme. • SHA384 and SHA512 support in X509 certificates processing. • Integration with ATECC508 (ECDSA and ECDHE support). • Supported cipher suites are: TLS_RSA_WITH_AES_128_CBC_SHA TLS_RSA_WITH_AES_128_CBC_SHA256 TLS_RSA_WITH_AES_256_CBC_SHA TLS_RSA_WITH_AES_256_CBC_SHA256 TLS_DHE_RSA_WITH_AES_128_CBC_SHA TLS_DHE_RSA_WITH_AES_128_CBC_SHA256 TLS_DHE_RSA_WITH_AES_256_CBC_SHA TLS_DHE_RSA_WITH_AES_256_CBC_SHA256 TLS_RSA_WITH_AES_128_GCM_SHA256 TLS_DHE_RSA_WITH_AES_128_GCM_SHA256 TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (requires ECC508) TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (requires ATECC508)	
Networking Protocols	
DHCPv4 (client/server) DNS Resolver IGMPv1, v2. SNTP	• DHCP client fixed to no longer send empty hostname (option 12)
Power saving Modes	
• M2M_PS_MANUAL • M2M_PS_DEEP_AUTOMATIC	No change
Device Over-The-Air (OTA) upgrade	
• Built-in OTA upgrade available. • Backwards compatible as far as 19.4.4, with the exception of: - Wi-Fi Direct (removed in 19.5.3) - Monitor mode (removed in 19.5.2)	• Fixed a bug where a particular OTA failure code-path could result in WINC1500 switching to an invalid image in the other partition. • Numerous other stability/robustness fixes
Wi-Fi credentials provisioning via built-in HTTP server	

Features in 19.7.10	Changes in 19.7.11
Built-in HTTP/HTTPS (TLS server mode) provisioning using AP mode (Open, or WPA/WPA2 secured).	No change
Ethernet Mode (TCP/IP Bypass)	
Allow WINC1500 to operate in WLAN MAC only mode and let the host send/receive Ethernet frames.	No change
ATE Test Mode	
Embedded ATE test mode for production line testing driven from the host MCU.	No change
Miscellaneous features	
	No change

3 Test Information

Please refer to ticket W1500-991 for full details.

Testing was performed against the release candidate 19.7.11 against the following configuration(s):

H/W Version : WINC1510 Xplained module

Host MCU : ATSAMD21-Xplained

The following testing was performed in both open air and shielded environments;

1. General functionality including:
    1. HTTP Provisioning
    2. Station Mode
    3. AP Mode
    4. IP (TCP and UDP client and server)
    5. HTTP POST/GET
    6. WPS (PIN and PushButton methods)
    7. Over-The-Air (OTA) update functionality and robustness (with and without TLS)
2. TLS functionality including:
    1. RSA cipher-suites:
        i. TLS_RSA_WITH_AES_128_CBC_SHA
        ii. TLS_RSA_WITH_AES_128_CBC_SHA256
        iii. TLS_RSA_WITH_AES_128_GCM_SHA256
        iv. TLS_DHE_RSA_WITH_AES_128_CBC_SHA
        v. TLS_DHE_RSA_WITH_AES_128_CBC_SHA256
        vi. TLS_DHE_RSA_WITH_AES_128_GCM_SHA256

        Testing uses 1024-bit, 2048-bit and 4096-bit server certificates, with a chain of 7 certificates of varying key lengths (1024,2048 and 4096 bit) leading to a 2048-bit root certificate.
    2. ECDSA ciphersuites:
        i. TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
        ii. TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

        Testing uses a NIST standard ECC P256 prime curve server certificate with two chains, one leading back to an ECDSA root certificate and the other leading to an RSA root certificate.
    3. Client authentication
3. Performance under interference
4. TCP/IP stack robustness testing
    1. Using an internal implementation of IPerf.
    2. Verification of multi socket functionality

4 Known issues

ID	Description
W1500-63	Occasionally WINC15x0 fails to receive an individual UDP broadcast frame when in M2M_PS_DEEP_AUTOMATIC powersave mode. Recommended workaround: **Use M2M_NO_PS powersave mode if reliability is preferred for UDP broadcast frames. Otherwise ensure the overlying protocol can handle the odd missing frame.**
W1500-108	The WINC15x0 cannot handle two simultaneous TLS handshakes, due to memory constraints. Recommended workaround: **When attempting to open two secure sockets in STA mode, the application should wait to be notified of the first one completing (succeeding or failing) before attempting the second one.**
W1500-325	1% of Enterprise conversations fail due to the WINC15x0 not sending an EAP response. The response is prepared and ready to send but does not appear on the air. After 10 seconds the firmware times-out the connection attempt and the application is notified of the failure to connect. Recommended workaround: Configure the authentication server to retry EAP requests (with interval < 10 seconds). **The application should retry the connection request when it is notified of the failure.**
W1500-369	When connected to certain access points, the WINC15x0 sometimes fails to roam when the access point changes channel. The issue is seen with these access points: Linksys E2500, Linksys E4200, Linksys 6500. The failures to roam are due to two issues: 1. Sometimes the access point takes a long time to start sending beacons or probe responses on the new channel, so it is not discoverable. 2. Sometimes the access point does not initiate the 4-way handshake (for WPA/WPA2 PSK reconnections). Recommended workaround: **On reception of M2M_WIFI_DISCONNECTED event, the application should attempt to discover the access point using `m2m_wifi_request_scan()` API.**

W1500-387	If an AP uses an 802.11 ACK policy of “No Ack”, then the WINC15x0 sometimes fails to receive 802.11b frames. Recommended workaround: **Avoid using an ACK policy of “No Ack”. If “No Ack” is used, ensure frames are sent at 802.11g or higher rates.**
W1500-397	70% of Enterprise connection requests fail with a TP Link Archer D2 access point (TPLink-AC750-D2). The access point does not forward the initial EAP Identity Response to the authentication server. The issue is bypassed by PMKSA caching (WPA2 only), so reconnection attempts will succeed. Recommended workaround: **The application should retry the connection request when it is notified of the failure.**
W1500-402	Occasionally during AP provisioning, after entering the credentials of the AP to connect to and pressing “connect”, an error will be returned even though provisioning was successful and the connection proceeds. Recommended workaround: **Add a delay in the application between receiving the provisioning info and connecting to the AP. Ignore the “Request Failed” message.**
W1500-699	When using a driver pre – 19.6.0 with this firmware, upon failure to obtain a DHCP address the WINC will not trigger a WiFi Disconnection and notify the driver of the failure. Recommended workaround: **In this case of an older driver running with later firmware, the application should monitor the time taken to obtain a DHCP address, if it takes too long then it can decide whether to disconnect and try again.**
W1500-854	When sustaining a maximum throughput TLS RX stream using a TLS record size of 16K, if another maximum throughput TLS RX stream with the same record size is established, the streams sometimes become stuck and fail to transfer any more data. Recommended workaround: **Try to avoid using 2 concurrent TLS streams receiving 16K records. If this scenario is used and the streams become stuck, close and reopen the sockets and try again.**
W1500-836	TLS Server mode does not work with ECDHE-ECDSA ciphersuites. Recommended workaround: **Ensure other ciphersuites are enabled before starting TLS Server. (The WINC15x0 firmware automatically disables ECDHE-ECDSA ciphersuites in TLS Server mode.)**
W1500-865	When running a sustained high throughput TCP RX stream over a long period (>1 hour), the WINC can sometimes miss enough beacons from the AP to trigger link loss detection and will disconnect from the AP. Recommended workaround: **If the WiFi connection is closed by the WINC during a TCP RX stream, reconnect and re-open the socket and try again.**

5 Fixes and enhancements

These are the major fixes and enhancements since the previous released version (19.7.10).

5.1 Issues fixed

ID	Description
W1500-327	**WPA2 Enterprise connection attempts sometimes take 30 seconds** When connecting to a WPA2 Enterprise access point, if the first EAP Identity Request frame from the AP was lost, the AP would retry the frame, but the WINC1500 duplicate detection logic would erroneously identify this as a replay and drop it, causing a delay until the next retry. Fixed: The duplicate detection lookup table is now initialized in a way that prevents false positive duplicate detection.
W1500-852	**TLS Server mode requires an ECDSA certificate chain in order to use ECDHE-RSA ciphersuites** When TLS Server mode was configured with ECDHE-RSA ciphersuites enabled, the server would fail to start unless an ECDSA certificate chain was present in flash. An ECDSA certificate chain should only be required for ECDHE-ECDSA ciphersuites, not for ECDHE-RSA. Fixed: TLS Server mode now correctly allows ECDHE-RSA ciphersuites without requiring an ECDSA certificate chain.
W1500-974	**OTA considered successful if no data is received** If an OTA download times out or is aborted before any data is received, the HTTP client code would still mark the download as successful. This could result in an empty or corrupt flash partition being marked as valid, and the host being incorrectly notified of OTA success If the host subsequently instructs the device to switch partitions (believing the OTA succeeded), the device may switch to an empty or corrupt image, causing device failure. Fixed: The OTA process now properly detects when no data has been received and reports a failure instead of success.
W1500-977	**DHCP client sends empty host name (option 12)** The WINC1500 internal DHCP client includes hostname option 12 in DHCP messages even when the hostname has been set to an empty string, resulting in an empty hostname field. This violates the DHCP specification which states that if option 12 is included, it must not be empty. Some DHCP servers reject DHCP requests with empty hostname options, preventing the WINC1500 from obtaining an IP address. Fixed: The DHCP client now only includes hostname option 12 when a non-empty hostname has been set via m2m_wifi_set_device_name()

W1500-978	**Override incorrectly programmed "MAC Address Used" bit in efuse** Some WINC1500 modules may have the MAC address programmed into efuse but the "MAC Address Used" flag bit incorrectly left unset. This could prevent the WINC1500 from using a valid MAC address that is present in efuse. Fixed: The firmware now validates the MAC address bytes directly and uses the MAC address if it appears valid, regardless of the "MAC Address Used" flag bit state.
W1500-979	**OTA doesn't report error if all sockets are in use** When all 7 sockets are already in use and an OTA is initiated, no error is reported via the host API. The OTA appears to start but makes no progress, and only an explicit abort command reveals that an OTA was in progress. Fixed: OTA now fails gracefully and reports an appropriate error if its socket is already being used by the host.
W1500-980	**OTA doesn't report further updates or error if the OTA socket is attempted to be opened after OTA is started** When an OTA is in progress (internally this uses socket 6) and the host attempts to open another socket which happens to also be socket 6, the socket connection fails as expected but the OTA process hangs without reporting progress or errors. The OTA appears active but makes no further progress. Fixed: The OTA socket (socket 6) is now blocked from use by the host while an OTA is active, preventing conflicts and ensuring proper OTA operation.
W1500-984	**TLS root certificate check is bypassed if server sends only a self-signed certificate** A critical security vulnerability was introduced in version 19.7.10 where the TLS root certificate validation was inadvertently bypassed when a server sends a certificate chain consisting of only a self-signed certificate. This was introduced as part of the certificate chain processing improvements made for W1500-929. This vulnerability allows an attacker with network access to spoof any TLS server by sending a self-signed certificate, as the WINC1500 would accept it without verifying against the trusted root certificates stored in flash. This issue only affects version 19.7.10; earlier versions (19.7.9 and prior) are not affected. Fixed: Restored proper root certificate verification for all certificate chains, including those containing only a self-signed certificate.
W1500-990	**Apply mandatory MISRA fixes to ASF driver** Mandatory MISRA C coding standard warnings were identified in the WINC ASF driver code. Fixed: All mandatory MISRA warnings in the ASF driver code have been resolved.

5.2 Enhancements

None

6 Appendix A – TLS Root certificates

The WINC1500 19.7.11 module comes with a preselected selection of TLS root certificates that will allow a TLS connection to be established with a range of internet TLS servers out of the box.

These preselected certificates are described in 6.1

6.1 TLS root certificates

Issuer	Filename	Expiry	Public Key	Signature Alg.	Notes
Amazon Root CA 1	AmazonRootCA1.cer	17 January 2038 01:00:00	RSA (2048 bits)	SHA256RSA	AWS Cloud
Baltimore CyberTrust Root	BaltimoreCyber-TrustRoot.cer	13 May 2025 00:59:00	RSA (2048 bits)	SHA1RSA	Azure Cloud
DigiCert High Assurance EV Root CA	DigiCert.cer	10 November 2031 01:00:00	RSA (2048 bits)	SHA1RSA	
DigiCert High Assurance EV Root CA	DigiCertSHA2.cer	22 October 2028 13:00:00	RSA (2048 bits)	SHA256RSA	
Entrust Root Certification Authority	EnTrust.cer	27 November 2026 21:53:42	RSA (2048 bits)	SHA1RSA	
GlobalSign Root CA	GlobalSignRoot.cer	28 January 2028 13:00:00	RSA (2048 bits)	SHA1RSA	
Internet Security Research Group Root X1	isrgrootx1.cer	04 June 2035 12:04:38	RSA (4096 bits)	SHA256RSA	LetsEncrypt
QuoVadis Root CA 2	QuoVadis_Root.cer	24 November 2031 19:23:33	RSA (4096 bits)	SHA1RSA	
VeriSign Class 3 Primary Certification Authority	VeriSign.cer	17 July 2036 00:59:59	RSA (2048 bits)	SHA1RSA	

7 Terms and Definitions

Term	Definition
AES	Advanced Encryption Standard
AJAX	Asynchronous JavaScript and XML
AKM	Authentication and Key Management
ARP	Address Resolution Protocol
ATE	Automated Test Equipment
BSS	Basic Service Set
CBC	Cyclic Block Chaining
DHCP	Dynamic Host Control Protocol
DHE	Diffie-Hellman Ephemeral
DNS	Domain Name Server
DTIM	Directed Traffic Indication Map
EAP	Extensible Authentication Protocol
EAPOL	EAP Over LAN
ECC	Elliptic Curve Cryptography
ECDHE	Elliptic Curve Diffie-Hellman Ephemeral
ECDSA	Elliptic Curve Digital Signature Algorithm
EEPROM	Electrically Erasable Programmable Read Only Memory
ESD	Electrostatic Discharge
EVM	Error Vector Magnitude
HIF	Host Interface
HTTP	Hypertext Transfer Protocol
IEEE	Institute of Electronic and Electrical Engineers
MAC	Media Access Control
OTA	Over The Air update
PEAP	Protected Extensible Authentication Protocol
PLL	Phase Locked Loop
PMK	Pair-wise Master Key
PSK	Pre-shared Key
QAM	Quadrature Amplitude Modulation
RSA	Rivest-Shamir-Adleman (public key cryptosystem)
RSN	Robust Security Network
RSSI	Receive Strength Signal Indicator
SHA	Secure Hash Algorithm
SNTP	Simple Network Time Protocol
SPI	Serial Peripheral Interface
SSID	Service Set Identifier
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TIM	Traffic Indication Map
TLS	Transport Layer Security

Term	Definition
WEP	Wired Equivalent Privacy
WINC	Wireless Network Controller
WLAN	Wireless Local Area Network
WMM™	Wi-Fi Multimedia
WMM-PS™	Wi-Fi Multimedia Power Save
WPA™	Wi-Fi Protected Access
WPA2™	Wi-Fi Protected Access 2 (same as IEEE 802.11i)